

MD.FARHAN BIN SHAFIQ

Full Stack Developer | Enterprise Web Applications | Agentic AI Workflows

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PROFESSIONAL SUMMARY

Full Stack Developer with 4+ years of experience working across the full stack, with a strong focus on secure, enterprise-grade applications and data-driven systems across healthcare and research platforms. At CT Health, I independently built a full-stack pharmaceutical ERP system handling inventory, billing, and procurement for a multi-role platform — and maintained the database for a nationwide diabetic patient registry. At ARCED Foundation, where I currently work, I developed a procurement and ticketing system with RBAC, JWT-based authentication, and automated approval workflows — reducing processing time by 60%+ — alongside a full enumerator management dashboard featuring multi-level approvals, biometric identity verification, and real-time field activity monitoring. I have hands-on experience designing PostgreSQL and MySQL schemas, building secure RESTful APIs with Node.js and Express, implementing role-based access control across complex multi-stakeholder platforms, and deploying containerized applications with Docker. I actively use spec-driven development and AI-assisted workflows to accelerate delivery while maintaining code reliability and clear technical documentation. I am currently expanding into Spring Boot to broaden my enterprise Java backend capabilities.

Employment History:

Web Developer, Arced Foundation

03/11/2023 — Continuing

- Developed research dashboards and data visualization tools using Python, Pandas, and Streamlit, including an enumerator management dashboard for attendance tracking, approvals, messaging, and field activity monitoring.
- Adopted agentic AI coding workflows (Claude Code, Cursor, OpenCode) with spec-driven planning and AI-assisted unit testing to accelerate feature development and maintain code quality across dashboard and automation projects.
- Built and integrated a face detection and verification system within the enumerator management workflow to validate user activities and improve operational monitoring.
- Developed custom SurveyCTO plug-ins and deployed ODK Central on a private server, building an automated data bridge with Google Sheets via REST APIs for real-time, centralized reporting.
- Implemented an offline voice-to-text solution using the Vosk AI model to support faster and more accurate survey data collection in low-connectivity environments.
- Built a full-stack procurement and ticketing system (React.js, Node.js, Express.js, PostgreSQL) with RBAC, JWT-based authentication, and automated approval workflows — reducing processing time by 60%+ for 50+ staff
- Developed and maintained the backend administration panel for the Arced Foundation website [🌐](#), enabling dynamic content management and operational control.
- Containerized and deployed the enumerator management dashboard using Docker, ensuring consistent, portable deployment across environments.

Web Application Developer, CT Health Ltd

01/01/2023 — 31/10/2023

- Independently developed a full-stack pharmaceutical ERP system covering inventory, billing, procurement, and reporting modules using React.js, Node.js, and MySQL.
- Managed and maintained the database for BNDR (Nationwide Electronic Registry) using MySQL, ensuring data integrity for nationwide diabetic patient records across multiple roles including admin, doctors, and operators.
- Designed and developed the frontend interface for CT Health Medicine, a multi-role pharmacy management application, using JavaScript, jQuery, and Bootstrap.
- Provided technical support for web-based systems to internal teams, ensuring system reliability for internal clinical teams.
- Managed and maintained the MySQL database, ensuring data integrity and consistent application functionality across multi-role pharmacy operations

Trainee Web Developer, Brand Solution Ltd.

15/12/2020 — 31/12/2022

- Gained hands-on experience with frontend development workflows, cross-browser debugging, and client-facing web delivery in a professional agency environment.
- Designed and maintained client websites, focusing on page speed and cross-browser compatibility.
- Worked alongside various teams to collect requirements and integrate new website features.
- Provided in-house technical support and troubleshooting for website-related issues.
- Assisted clients with website inquiries and support, ensuring timely resolution of their needs.

TECHNICAL SKILLS

- **Full Stack Web Development:** React.js (Hooks, Context API, React Router), Node.js, Express.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap, jQuery, RESTful API Design & Security, JWT / OAuth2, Webpack, NPM
- **Data Science & Visualization:** Python (Pandas, Matplotlib, Streamlit), Stata
- **Data Engineering:** ETL / Data Pipeline Development, REST API Integration, Data Synchronization
- **AI & Automation:** AI Agents, Agentic Coding Workflows (Claude Code, Cursor, OpenCode), Spec-Driven Development, AI-Assisted Unit Testing, Face Detection & Biometric Verification (InsightFace), Offline Voice-to-Text (Vosk), Google Apps Script, SurveyCTO
- **Database Management:** MySQL, PostgreSQL (Schema Design, Query Optimization, Indexing), MongoDB, Firebase
- **Deployment & Tools:** GitHub, Docker (Containerization & Deployment), Server Management, CI/CD (GitHub Actions), Netlify, Scrum (Agile)

Education

Bachelor of Science (BSc) in Computer Science & Engineering, North South University

Higher Secondary Certificate (HSC), Cambrian School and College

Secondary School Certificate (SSC), Siraj Uddin Sarker Vidyaniketan and College

Projects

Enumerator Management Dashboard

- Architected and deployed a secure, role-based enumerator management dashboard using Python Streamlit to centralize operations, streamline multi-level approval workflows, and automate internal communication processes.
- Developed a dynamic nomination and tracking module that enabled project managers to efficiently search, filter, and assign verified enumerators based on experience, education, skill set, and geographic location.
- Implemented a biometric verification system using InsightFace for high-accuracy face detection, facial alignment, and identity validation of enumerator profile images.
- Built a multi-layer anti-spoofing and liveness detection pipeline using Local Binary Pattern (LBP), color space analysis, and FFT-based Moiré pattern detection to reduce screen replay and paper-based identity fraud.
- Optimized large-scale data processing workflows using Pandas and MongoDB, integrating duplicate profile detection through facial embeddings to maintain data integrity, platform reliability, and operational security.
- Containerized the application using Docker for reliable deployment and environment consistency

ARCED Forge – Internal Procurement & Ticketing System

- Architected and delivered a full-stack internal procurement and ticketing platform for 50+ staff — covering end-to-end development from database schema design to frontend implementation using React.js, Node.js, Express.js, and PostgreSQL.
- Implemented Role-Based Access Control (RBAC) with a multi-tier approval hierarchy across staff, logistics, and senior management, enforcing strict data visibility and permissions per role.
- Built secure RESTful APIs with JWT-based session management to handle requisition workflows, real-time approval tracking, and fulfillment notifications.
- Developed a digital requisition module enabling staff to raise requests, track approval status in real time, and receive fulfillment updates — replacing manual email chains entirely.
- Automated the approval workflow engine, reducing processing time by 60%+, and built an audit trail module for on-demand transaction logs and procurement reports.
- Designed and optimized the PostgreSQL schema to support RBAC, approval hierarchies, and full transaction logging.
- Containerized and deployed the application using Docker for consistent, portable deployment across environments.

CT Health Medicine – Pharmaceutical ERP System

- Designed and developed a full-stack multi-role pharmaceutical ERP system independently, handling both frontend and backend across all modules — from database design to UI implementation.
- Built a role-based access control system ensuring admins, stock managers, and outlet heads each had appropriate data visibility and operational permissions across the platform.
- Developed a real-time inventory management module tracking medicine stock levels, purchase sources, supplier records, and units sold — maintaining accurate stock data across all outlets at all times.
- Implemented an automated billing system that generated invoices at point-of-sale, eliminating manual billing processes and reducing transaction errors.
- Built a procurement and requisition module allowing outlet heads to raise purchase requests, track approval status, and maintain complete supplier-wise purchase history.
- Developed a reporting module enabling management to generate procurement summaries, sales breakdowns, and inventory movement reports on demand — reducing manual reporting effort significantly.

BNDR – Nationwide Diabetic Registry System

- Managed and maintained the MySQL database for a nationwide electronic health registry serving diabetic patients across Bangladesh, supporting multi-role access for admins, doctors, and operators.
- Ensured data integrity and consistency across patient records, visit histories, and prescription image uploads through structured database maintenance and query optimization.
- Collaborated with the development team to support backend data operations and system reliability for a high-volume, multi-stakeholder healthcare platform.

Custom SurveyCTO Plug-ins & ODK Integration

- Developed custom SurveyCTO field plug-ins to enhance survey UI and data collection workflows.
- Deployed ODK Central on a private Linux server and built an automated data bridge integrating it with Google Sheets via REST APIs for real-time, centralized data synchronization.
- Reduced manual data handling errors and enabled faster reporting for field survey operations.

Offline Voice-to-Text Survey System

- Implemented an offline speech-to-text system using the **Vosk AI model** for survey data collection.
- Enabled accurate voice input in low-connectivity environments, improving enumerator productivity.
- Integrated the system with existing survey workflows for seamless data capture.

LEADERSHIP & VOLUNTEERING

Joint Secretary, NSU Computer & Engineering Club

2017 – 2018

- **Nationwide Event Management:** Orchestrated the "Cybernauts Programming Contest," managing over 3,000+ participants. Led the entire event lifecycle from infrastructure setup and question curation to successful execution.
- **Team Leadership:** Served as Team Lead for the University Convocation and organized major Robotics Competitions, ensuring seamless logistics and volunteer coordination.

General Member, JCI Dhaka Aspire

- **Achievement:** Awarded "**Quarterly Best Member**" for consistent contribution and active participation in community projects.